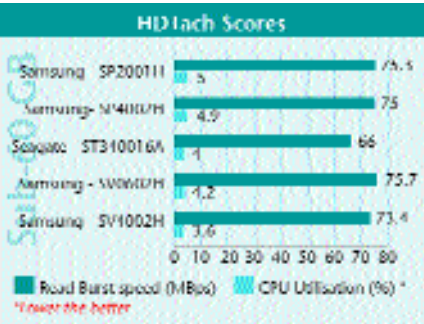


idea to ensure that the cabinet you use this drive in has good cooling and airflow.

Performance

We tested the drives for their performance using synthetic benchmarks as well as for their real-world performance characteristics. We used the traditional drive benchmarking tools and software for obtaining a drive’s raw speed in terms of data transfers and access time. The access time was logged using real world applications such as *Return to Castle Wolfenstein* and Photoshop 7.0. The overall score was based on a weighted average of these two sets of scores.

HDTach 2.6: This is a synthetic test which records the read burst speed and CPU utilisation. The program calculates the total CPU usage of the drive and the burst transfer rate, which



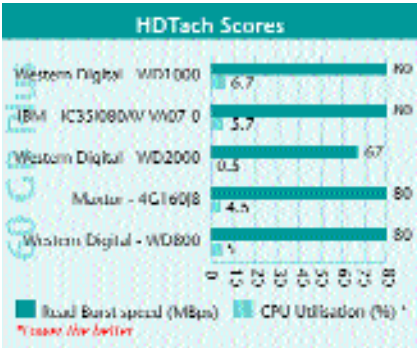
is a measure of the maximum possible bandwidth of the drive over its interface.

Up to 60 GB capacity: The read burst speed is a measure of the maximum transfer rate that the drive can support over its interface and is there-

fore directly related to the IDE interface the drive is using, coupled with its spindle speed. Six of the eight drives managed to hit read burst scores of above 70 MBps. However, both the 40 GB drives from Seagate (the ST340016A and the ST340810A) couldn’t make it past this level. The one drive that really shone in this test was the Maxtor 541DX—it was the only drive in this category to log a data transfer rate of 80 MBps.

At 3.1 per cent, the CPU utilisation of the Maxtor 541DX was the second lowest in this category—the drive with the lowest CPU utilisation was the 40 GB Seagate ST340810A drive with a score of 2.7 per cent. The low CPU utilisation of both these drives makes them suited to applications where you would multitask many applications on your PC such as downloading applications over your broadband connection while watching a DivX movie off your hard disk. This also implies that these drives can be used in older computers with slower processors. As they do not strain the processor much, they would give a smoother operation. The drive that took a beating here was the Samsung SP2001H—it logged a CPU utilisation of 5 per cent.

Over 60 GB capacity: In this category, the lone IBM drive, both the Maxtor drives, and two of the three Western Digital WD1000 and WD800 showed an



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